TRACE: Certified Mealy-to-Puzzle Construction for Temporal Causal Explanation

Pre-trial instrument manuscript. This document describes the v5 general machine-to-puzzle package and target-display correction for the finite explanation task. Computational checks and synthetic sessions are not human or language-model findings. Orange result slots remain until corresponding evidence is collected and analyzed. The separate public lab demo adds continuous local feedback; it is not an authenticated collection system or an implementation of a completed human trial.

Abstract

We present TRACE, a benchmark of temporal causal explanation in fully specified finite-state systems. Given a total deterministic Mealy machine, a valid observed trace and a target output at a finite time, a deterministic compiler constructs a switch-and-light puzzle. The executable semantic constructor is proved in Lean to preserve every output round under every input intervention, including machines with loops and outputs that turn off after turning on. A separately proved terminal projection removes post-target state distinctions from the display while preserving the scored event under every intervention. Given the recorded positive target event, a participant selects one inclusion-minimal set of recorded input values that both guarantees the event when fixed and permits prevention by changes confined to that set. This contract distinguishes explanatory minimality from minimum-cardinality optimization and accepts every valid minimal alternative. We construct a bounded two-player verification game whose winning condition is exactly the combined causal predicate. Backward evaluation computes its two branches; an absorbing-sink encoding relates the finite verdict to minimum-even parity acceptance. Upward closure yields a terminating deletion algorithm for finding one minimal explanation. Local recurrence certificates support positive and absence claims. A standalone dataset constructor is optional: external machine, trace and target records enter the same compilation boundary. The study design uses 1,024 agent-test items and a separate 64-item shared human–agent panel. A common session service provides the same network, recording, teaching examples, intervention budget, and answer contract to both populations. The browser teaches the task through animated examples; formal transition tables remain an optional reference. Human ease, comparative agent performance, and generalization remain empirical questions.

RESULTS REQUIRED — TRIAL-PRIMARY-ABSTRACT

Insert actual analyzed human and agent samples, registered configurations, matched-panel exact accuracy and justified uncertainty. State the strongest supported finding without importing full-pool agent scores into the 64-item comparison.

1 Question and scope

A known transition system permits an exact distinction between predicting its behavior and explaining a recorded event. TRACE asks which recorded inputs jointly provide a guarantee and a means of preventing that event. The complete rules are public; the task does not require learning a hidden rule from examples.

TempoBench motivates the separation of forward simulation and temporal causal attribution over deterministic Mealy executions [3]. Its version-3 fixing and supported-change conditions motivate our explicit contract. We do not claim reproduction of its released labels, CORP pipeline, prompts, or scores. Reasoning alignment motivates the task; file-format and gold-label compatibility are separate claims that we do not make. The dataset constructor emits machines and recordings independently of the game. The compiler imports those records without regenerating their machine or selecting another trace. The intended human–AI comparison shares ARC's interest in comparing populations, but does not copy its grid tasks, hidden-rule setting, or measure of general intelligence.

Broader reactive-system cause synthesis uses automata to represent temporal causes and effects [1]. This paper makes no equivalence claim between those cause languages and its finite cell sets.

The reverse-parity review underlying this project separates nearest-removal trace necessity from strategy-guarantee necessity. Neither is silently identified with the combined finite predicate developed here. Its nearest-removal literal-conjunction specialization yields pivotal singletons under unrestricted Boolean edits. The strategy question instead requires an arena of alternative system moves and asks about one policy's universal guarantee together with its existential violating behavior. Our fixed-machine input interventions are a different, explicitly defined task.

This bounded verification game does not recover an original synthesis specification or arena. Time runs forward; the algorithm evaluates possible continuations backward from the designated event. Arbitrary infinite liveness and policy replacement are outside this release.

2 Machine and intervention model

Let

$$M = (Q, q_0, \Sigma, \{0,1\}, \delta, \lambda), \qquad \Sigma = \{0,1\}^m,$$

be a total deterministic Mealy machine. Its run on inputs $x_0, \ldots, x_k$ satisfies

$$q_{t+1} = \delta(q_t, x_t), \qquad o_t = \lambda(q_t, x_t).$$

Output uses the pre-transition state and current input. Let $h = k+1$, $D = \{0, \ldots, k\} \times \{0, \ldots, m-1\}$, and let a be the recorded assignment to D. Define $E(x)$ by $o_k(x) = 1$ and require $E(a)$. Every Boolean assignment to D is admissible. The machine and initial state remain fixed during intervention; outputs and internal states are recomputed.

The input contract imposes no fixed maximum state, channel or horizon count. Every supplied instance is finite. Machine cycles are allowed because the target is evaluated after a finite prefix; infinite-horizon strategy solving is not needed. The initial contract fixes the target to output 1 at step k and admits all Boolean input assignments. Environmental constraints and richer event predicates require a separately specified and verified extension.

A cell identifies its recorded value, including zero. The serialized cell ID is $tm + r$, and the row index of a valuation is $\sum_r x_r 2^r$. These conventions are public in both modalities. Mealy prefix locality ensures that inputs after k cannot affect this target.

Definition 1 (Combined finite explanation)**.** *For $C \subseteq D$, define*

$$G(C) \iff \forall x\, (x|_C = a|_C \Rightarrow E(x)), \qquad P(C) \iff \exists x\, (x|_{D\setminus C} = a|_{D\setminus C} \wedge \neg E(x)).$$

Write $H(C) = G(C) \wedge P(C)$. An accepted explanation is a set C for which $H(C)$ holds and no proper subset satisfies H.

Prevention may change only some selected cells. It is not the assertion that flipping every selected cell defeats the outcome. XOR at $(1, 0)$ has the two-cell combined explanation, although flipping both preserves the target. For OR at $(1, 1)$, either singleton guarantees the output but neither permits prevention; the pair satisfies both. For AND at $(1, 1)$, either singleton permits prevention but neither guarantees the output; again the pair is required.

There can be multiple accepted answers of different sizes. Minimality refers to inclusion, not global minimum cardinality. The participant submits one accepted explanation, not the family or its union. A separate `no_explanation` verdict is correct when no valid set exists. An empty cell list does not encode that verdict.

3 A general machine-to-puzzle construction

Dataset construction, benchmark compilation and participant interaction are separate components. The optional constructor supplies machine/trace/target records. The compiler validates the imported machine and complete observed run, retains the prefix through the target, and creates one common task. Human and agent interfaces read this task and use a shared session service and grader. No switch-network recipe is required to import an external machine.

For each time $t \leq k$ and state $q \in Q$, construct a binary decision tree that reads the m input channels in order. The decision for channel r refers to the original cell $tm + r$, and its two branches represent that cell's values. The leaf for valuation u emits $\lambda(q, u)$ and points to state $\delta(q, u)$ in the next time layer. Traversing a tree is one semantic round, regardless of the number of switch decisions. Thus outputs can change every round; an earlier positive output is not an absorbing success state.

All appearances of a given time/channel cell refer to the same input setting and explanation mark. Routing nodes introduce no new interventions. Identical subtrees may be merged by a proved equality reduction, while the original full cell inventory remains available. With an explicit transition table, the full unreduced forest has $O(h|Q|2^m)$ nodes. This representation is finite, but can still be large; equality reduction is not a claim of universal compression.

Theorem 1 (General construction correspondence)**.** *For every compatible total deterministic Mealy machine, every finite target k, every input word x through that target and every $t \leq k$, the compiled puzzle emits exactly $o_t(x)$. The original and compiled tasks have the same guarantee and prevention predicates and exactly the same inclusion-minimal explanations. These conclusions also hold after identical-branch reduction.*

Proof. Induction on the number of input channels shows that a decision tree reaches exactly the leaf for its supplied valuation. That leaf preserves both the emitted output and successor state. Induction on rounds then preserves every output and state transition. Equality reduction preserves tree evaluation because choosing between equal subtrees changes no result. Cell IDs are injective in time/channel pairs; the assignment domain and actual values remain unchanged. Replacing the effect by an extensionally equal effect therefore preserves both quantifiers, their conjunction and inclusion minimality. □

The compiler CLI calls the executable `compileRoundReduced` definition proved in Lean. Structural recursion establishes termination for each finite tree and replay prefix. Deterministic channel order, state order and equality reduction fix the semantic artifact for a given canonical input and compiler version. Canonicalization under arbitrary renaming or machine isomorphism is not claimed.

The semantic forest retains the complete transition pair at every round, including the target. The final visual decision tree instead observes only its output: map each terminal leaf $(\delta(q,u),\lambda(q,u))$ to $\lambda(q,u)$ and reduce equal branches again. Denote this projection by $\Pi(T) = \mathrm{reduce}(\mathrm{map}_{\mathrm{snd}}\, T)$. All true leaves lead to one shared target lamp; false leaves lead to a distinct exit that leaves the target dark. Intermediate outputs are status indicators, visually distinct from the goal. A final-round control can disappear from a particular route when only the post-target state varies; it can remain relevant on another route, and its original cell stays in the intervention inventory.

Theorem 2 (Terminal display correspondence)**.** *For every semantic switch tree T and input assignment x,* $\mathrm{read}(\Pi(T), x) = \mathrm{snd}(\mathrm{read}(T, x))$. *Following the unchanged semantic transitions before k and the projected tree at k therefore returns exactly $o_k(x)$ for every x. The resulting task has the same G, P and inclusion-minimal explanations as the source machine.*

Proof. Induction on the tree shows that mapping its leaves commutes with evaluation. Equality reduction preserves that evaluation. Before the target, successor states are unchanged by Theorem 1; at the target the projection returns precisely the transition output. Extensional target equality with the unchanged input and candidate domains preserves the quantified predicates and minimality. The semantic forest still preserves the complete state sequence; the output-only terminal display deliberately does not encode the final state. □

The CLI uses the proved executable `compileTerminal` definition for this projection. Applying the same output-only reduction to earlier rounds would be unsound in general: equal current outputs can lead to different future outputs. Serialization of the projected tree and its depiction as one shared visual goal remain separately tested application obligations.

4 Backward evaluation and the verification game

Define

$$\mathrm{Bad}(F) \iff \exists x\ (x|_F = a|_F \wedge \neg E(x)).$$

Then $G(C) = \neg\,\mathrm{Bad}(C)$ and $P(C) = \mathrm{Bad}(D \setminus C)$. Let $A_F(t)$ contain exactly the input valuations respecting fixed cells in round t. Compute backward:

$$R_F(t,q) = \begin{cases} \bigvee\limits_{u \in A_F(k)} [\lambda(q,u) = 0], & t = k, \\ \bigvee\limits_{u \in A_F(t)} R_F(t+1, \delta(q,u)), & t < k. \end{cases}$$

Theorem 3 (Replay correspondence)**.** *$R_F(t,q)$ is true exactly when some permitted input suffix from pre-round state q defeats the target at round k. In particular,* $\mathrm{Bad}(F) = R_F(0, q_0)$.

Proof. Induct on remaining rounds. At the target, the recurrence tests exactly the permitted current-input outputs. Before the target, every permitted suffix consists of a permitted valuation followed by a permitted suffix from its deterministic successor. The disjunction and induction hypothesis give both directions. The terminal test retains the transition output; final state alone is insufficient. □

Definition 2 (Two-branch verification game)**.** *Fix C before play. At the root a challenger selects a branch. In the guarantee branch, the challenger supplies inputs outside C, values in C stay*

recorded, and the explainer wins iff E. In the prevention branch, the explainer supplies inputs inside C, values outside C stay recorded, and the explainer wins iff $\neg E$. The machine transitions are forced.

Theorem 4 (Game correspondence)**.** *The explainer wins from the root iff $H(C)$.*

Proof. Winning the guarantee branch requires success against every permitted completion, exactly $G(C)$. Winning prevention requires one permitted defeating completion, exactly $P(C)$. The challenger selects the branch after C is fixed, so both are required. The candidate is not selected adaptively after counterfactual input. □

Attach a priority-0 absorbing sink after a successful terminal transition and a priority-1 absorbing sink after failure. Every play reaches its sink in bounded time. Its least infinitely recurring priority is even iff the terminal verdict is successful. This is an exact parity encoding of the bounded game; general infinite parity solving is unnecessary. After the branch choice, only one player has meaningful choices in each branch.

5 Minimality and an executable algorithm

Theorem 5 (Upward closure and local minimality)**.** *Both G and P are upward-closed in C. A set is an accepted explanation iff $H(C)$ and $H(C \setminus \{c\})$ fails for every $c \in C$.*

Proof. Adding fixed cells restricts completions in G, preserving a guarantee. Adding editable cells enlarges possible witnesses in P, preserving prevention. If a valid proper subset $A \subset C$ existed, choose $c \in C \setminus A$; upward closure would make $C \setminus \{c\}$ valid. The converse follows from inclusion minimality. □

Start with $C = D$. If $H(D)$ fails, return no explanation. Otherwise consider every cell once, deleting it when validity survives. A failed deletion stays failed after subsequent deletions: a smaller valid set would imply validity of the earlier superset. Thus the procedure terminates after at most $|D|$ deletion decisions, preserves validity, and returns an inclusion-minimal answer by Theorem 5. Since $E(a)$, $G(D)$ is true. An explanation therefore exists iff some input assignment defeats the target; constant-true behavior has none.

A query visits $O(h|Q||\Sigma|)$ recurrence transitions once permitted valuations are available. Python's current fixing filter adds $O(hm|\Sigma|)$ work; concrete valuation encoding and certificate adapters have further bit-processing costs. Finding one answer uses a linear number of validity queries. The recurrence-work bound is $O(|D|h|Q||\Sigma|)$, not a precise runtime bound for every adapter. This counts $|\Sigma| = 2^m$; it is not a lower bound or a claim for minimum-cardinality search, all-answer enumeration, or general parity games. The ordinary decision path retains two state rows per query and caches only decision bits within a solver invocation. Detailed tables are constructed when certification or witness extraction requires them. Enumerating every accepted answer can require exponential output and is an explicit small-instance audit, not a dependency of ordinary construction or grading. Resource exhaustion is an incomplete computation, never evidence that the input is mathematically invalid or has no explanation.

Proposition 1 (Independent support characterization)**.** *Let J be the inclusion-minimal supports of defeating changes from a, and T the inclusion-minimal sets intersecting every member of J. The accepted family is*

$$\mathrm{Min}_{\subseteq}\{t \cup j : t \in T,\ j \in J\}.$$

Proof. Every defeating support contains a minimal defeating support. Thus $G(C)$ holds iff C intersects all members of J, and $P(C)$ iff C contains some member. Every valid C contains a minimal hitting set t and a member j, whose union remains valid. Minimality forces equality. Conversely, a valid proper subset of a minimal such union would contain a smaller such union, a contradiction. If J is empty, prevention fails everywhere and the family is empty. □

This proof needs no monotonicity of the machine output in its inputs. The paper proof and exhaustive tests of this characterization are distinguished from the machine-checked theorem inventory.

6 Certificates and formal coverage

A certificate binds the public instance and semantic version to the actual run, a proposed answer, and the query tables needed for guarantee, prevention and each deletion. The checker verifies dimensions, domains, every terminal row and every backward recurrence equation over all machine states. A true initial query has a permitted defeating input word. A false query is supported by the complete locally checked table. One successful path cannot establish a universal guarantee or absence claim.

The checker does not trust the producer's search order or chosen answer. Minimality follows from checked queries and Theorem 5. A no-answer certificate checks the globally free query. One canonical answer per compiled item is checked during package construction; grading still accepts all minimal alternatives by evaluating the submitted candidate directly. Enumerating and certifying every accepted answer is optional deep-audit work. Certification is mandatory for publishing a certified package; retaining detailed certificate records is optional. Verbose console diagnostics do not alter decisions, and a separate output option saves checked certificates. A compact manifest binds the package, source, compiler and validation identities.

The Lean development under `formal/` builds on reviewed Mealy uniqueness and locality definitions, with vendored-source provenance. New modules separate combined semantics, terminating deletion, layered recurrence and certificate checking, finite cell binding, general switch construction, equality reduction and terminal display projection, sparse checked-query lookup, and game/parity correspondence. The exact coverage and executable checks are recorded in `formal/README.md` and generated verification artifacts. A theorem count alone is not evidence that the formal statements match the task.

The semantic construction and terminal projection have the general theorems above; the SVG renderer does not. A proof that an output-only tree evaluates correctly does not prove that all successful paths are drawn toward the same recognizable goal. Canonical cell-key injectivity, insertion of locally checked roots, lookup binding and required-query coverage are also proved. The certificate runtime uses these exact lookup and coverage functions, eliminating a redundant recursive evaluation of the reference oracle after table checking. The remaining JSON parsing, finite-array assembly, serialization, imperative adapter control flow, source/digest binding, executable platform and web/session implementation are explicit tested trust boundaries. Independent per-transition checks validate every source valuation at every compiled state/round, including the terminal projection, without enumerating all complete words. Exhaustive small examples, direct quantified oracles, cross-language checks and deliberate mutations address adapter mismatches. Neither finite tests nor theorem counts prove human readability. No unrestricted LTL compiler or policy-solving correctness is claimed.

7 Dataset, difficulty and release identity

The standalone constructor produces machines, observed traces and targets before any game construction. The v5 package recompiles the same source machines, traces, targets, split assignments and panel membership as v4; its display correction does not resample the dataset or change accepted explanations. A fixed seed determines the producer's output; imported datasets use the same compiler without passing through this producer. The study package preserves the planned 16 tutorial, 32 pilot, 256 development and 1,024 test roles, with a separate 64-item matched human–agent panel. Exact released counts, hashes and executed checks belong to the generated validation block below and the manifest.

The constructor samples explicit total deterministic tables and positive actual executions. For temporally dependent selection it checks that fixing the final round alone does not guarantee the target, which implies that every valid explanation contains an earlier cell. Such filters shape the sampled task distribution; they do not change the general import contract. Rejection of exact machine duplicates is not semantic deduplication or evidence of held-out causal mechanisms. Any stronger split equivalence, leakage control or generalization claim needs its own declared test and evidence.

State count, input width and trace length define structural sampling strata, not a validated ordinal difficulty scale. The meaningful quantities also include reachable alternatives, temporal dependence, irrelevant cells, valid-answer multiplicity and presentation demands. Adding unreachable states or unused rounds can enlarge an instance without making its reasoning harder. TempoBench's structural performance analysis provides hypotheses for calibration, not thresholds already validated for this interface and answer contract. Human accuracy, agent accuracy, error type and navigation/time measurements must establish any empirical scale. The extended agent range need not remain solvable by humans; conclusions about a human limit require measured human evidence.

Dataset version `trace-mealy-v5` and protocol version `trace-study-v5` retain the normative `trace.gp.v2` predicate. The v4 presentation retained separate terminal bulbs for transitions with the same output but different successor states. This could suggest that reaching one particular bulb was the goal, although scoring used output 1 at the target time. V5 distinguishes the semantic transition from its terminal display and uses one shared target. V4 source, data and evidence are preserved under `archive/v4-multiple-targets`, and its raw playtest logs remain private. The prior v3 source, dataset and evidence are frozen under `archive/v3-switchlight`; earlier v2 material remains separately preserved. Historical observations retain their original release identity and cannot be silently pooled with the new instrument. Detailed generated certificate bundles and audit products are separated from the small production core. Raw trial records remain private and are not disposable build products.

Executed v5 and demo computational validation. The clean integrated demo release run passes 100 Python tests and the Lean build of 12 modules with 126 named theorems. These counts describe software checks and the formal inventory, not participant observations or independent empirical replications. The general construction theorem applies to all compatible finite inputs; JSON adapters, native execution and presentation remain tested trust boundaries.

All 1328 released items pass local source-transition correspondence, including all 23,360 terminal state/input output cases, and Python/Lean checking of one canonical explanation certificate each. Complete answer inventories are not required by compilation or scoring. A bounded independent audit checks 22,624 complete input words and every candidate subset for 368 items with at most eight cells. The actual browser semantic and terminal-display interpreters separately agree

with independent table execution on 514,144 exhaustive words through twelve cells, plus 1,280 deterministic probes on larger items. These counts do not claim exhaustive coverage of larger input words or answer families. Real-browser tests separately check the reported four-setting target regression, practice removal illustrations, shared endpoints, navigation and the external, larger and constant-output fixtures. These are synthetic checks, not evidence of human comprehension.

Two complete current compilations reproduce byte-for-byte. The source records are byte-identical to v4, whose standalone constructor reproduction is preserved as historical evidence. All original machines, traces, targets, semantic forests and forms remain unchanged. Every release item passes the earlier-round-dependence filter. Structural sampling ranges and this filter do not establish empirical difficulty or unseen-mechanism generalization.

Dataset identity. `77d46037ce07703f82db9a5076c7e493`
`313e4742eefa4571859c17ae3ac605bb`

Measured non-learning instrument baselines. The following table uses the fixed 1,024-item agent-test pool. Uniform subset sampling has five repetitions per item; other procedures have one. The resulting 10,240 records are computational instrument checks, not recruited-human or language-model outcomes. Percentages are observed rates; analytic chance is not computed by the production runner.

Non-learning baseline	Exact (%)	G (%)	P (%)	Minimal (%)
Uniform input subset	0.898	9.531	89.238	0.898
All input cells	0.781	100.000	100.000	0.781
No cells	0.000	0.000	0.000	0.000
Latest round	0.000	0.000	93.359	0.000
Single-cell pivots	26.660	26.660	97.559	26.660
Finite-game solver	100.000	100.000	100.000	100.000

The exact solver accepts any inclusion-minimal G/P explanation; its success is a computational consistency check, not an estimate of unaided reasoning. Full-pool baselines cannot be substituted for a matched-panel human–agent contrast. Detailed records remain in private generated artifacts, and compact reports bind their hashes and implementation identities.

8 Playable interface and evaluation

The board exposes the compiled switch routes, outputs and supplied recording. Each time layer has a mechanism for every state; navigation can focus a region without changing the semantic graph. Clicking a switch marks its recorded setting without changing that setting. Repeated appearances of one cell share one mark. The task is to mark switches that pass two tests: keeping the marked settings must force the lamp on at the designated target round, and changing only marked settings must permit it to be off at that round. All successful terminal routes join the same target lamp; the other exit leaves it dark. Intermediate output indicators may switch between ON and OFF and are not goals. No mark may be removable while retaining both properties. A replay shows the supplied recording without revealing new counterfactuals.

Five untimed animated demonstrations introduce a direct path, conjunction, disjunction, XOR and a removable extra mark. Demonstrations enumerate all visible completions for each obligation and distinguish a failed guarantee from failed prevention. A single successful animation is never presented as proof of a universal claim. Practice feedback provides the appropriate counterexample

or prevention witness, a certified example answer and redundant-mark feedback. Where appropriate, the latter compares the recorded setting with a changed setting to illustrate why a redundant control can lead to the same target. Such a pair is illustrative, not a complete enumeration or proof of the universal guarantee; the exact checker establishes the feedback verdict. Formal instructions, cell IDs and the equivalent transition table are available in an optional reference rather than required for viewing the puzzle.

8.1 Public learning-enabled demonstration

The separately versioned `trace.demo.v1` application exports the same certified tasks for static hosting on GitHub Pages. It provides immediate obligation feedback, explicit retries and solution disclosure, saved local progress, and continued play through the available finite pool. The first submitted answer remains separate from later attempts and eventual completion. An answer following earlier teaching or feedback measures performance after that exposure, not untrained performance. Replayed tasks are not counted as newly encountered tasks.

A JavaScript implementation computes the exact backward recurrence and deletion test without stored scoring keys. It is a separately tested implementation, not a newly kernel-proved extraction. Feedback witnesses illustrate the relevant quantifier; the full recurrence decides universal claims. The browser and local agent transport share task data, teaching, feedback and retained session history. This transport is not a participant sandbox, and source-accessible solvers cannot establish unaided performance. Browser/process-local progress is editable and is not an authenticated research record. No central score collection is provided by the static demonstration.

Structural navigation bands describe state count, input count and horizon; they do not establish measured difficulty. Calibration must distinguish causal structure from display burden and assess temporal dependence, conditional relevance, interacting inputs and redundant alternatives. Valid learned general strategies are legitimate success; incidental dataset shortcuts require contrasting tasks and mechanism-grouped transfer studies. Published v5 items are public exposures, irrespective of their historical split labels. Concealed-item claims require separately reserved material.

Hosted collection with durable first-submission records, retained feedback history, declared agent context/resource limits and exposure-aware analysis is a second milestone. It requires a newly specified learning-enabled protocol; the fixed-design study below cannot be reused unchanged for adaptive longitudinal records.

8.2 Preserved server-based study protocol

The following settings describe `trace-study-v5`, which remains separate from the public learning-enabled demo. They are a prepared study configuration, not reports of collected human or model observations.

Agents receive the same semantic graph, terminal display projection, cell identities, recorded settings, examples and practice feedback in structured public serialization. Humans can also inspect the equivalent serialization. Shared facts and budgets provide an information contract, not perceptual equivalence: visual navigation and structured-graph processing may differ. A matched image-only agent condition has not been implemented or evaluated. Recipes, certificates, answer families for scored items, and earlier test outcomes are withheld by the common session service.

The default reasoning condition permits no counterfactual probes. The optional study arm permits eight common simulator queries. The default study configuration allows 480 seconds and one scored submission. These are declared protocol parameters, not limits of the mathematical constructor; a different budget requires a distinct disclosed evaluation configuration. Practice

shares certified examples and feedback; scored trials provide receipt and status only. Replaying the supplied recording adds no counterfactual evidence.

Each human receives two practice items and four scored items with breaks between. Saved balanced allocation controls forms and rotations. The optional probe arm is between participants. Each agent item starts in a fresh context with a declared configuration. A coding assistant with repository access is not a participant.

Truncated provider responses are invalid even when partial text contains parseable JSON. Returned model identities are checked against declarations. Request identities, duplicate/rejected requests, withdrawals, deadlines and source/protocol identities are retained. A scored response cannot be replaced.

RESULTS REQUIRED — TRIAL-PILOT

Report the actual consenting pilot sample, comprehension, accessibility and timing issues, revisions and lock. Complete the institutional and participant-material records before recruitment.

RESULTS REQUIRED — TRIAL-SAMPLE

Report actual enrollment/disposition, item coverage, configurations, independent clusters, registration identity, source/release hashes and deviations.

9 Analysis and empirical slots

The primary comparative statistic is equal-item exact explanation accuracy on the fixed 64-item panel. Full-pool agent results use 1,024 items. The registered contrast is agent minus human accuracy. Every accepted minimal answer earns full credit. Cell overlap against a canonical answer is retired.

Report guarantee failure, prevention failure and valid-but-nonminimal responses alongside malformed outputs, timeouts, exclusions and withdrawals. Latency and probes need denominators. Answer shapes do not establish internal reasoning strategies. There is no scored simulation control in this release, so it cannot establish a measured SIM/MIN gap.

Humans are participant clusters; an agent run/repetition stays a cluster across all its forms. Under an outcome-independent fixed observation design, cluster c has contribution range width

$$w_c = \sum_{j \text{ observed by } c} \frac{1}{|J| n_j}.$$

Independence between clusters gives the conservative Hoeffding radius [2]

$$\epsilon_\alpha = \sqrt{\tfrac{1}{2} \log(2/\alpha) \sum_c w_c^2}.$$

Combine squared weights for independent population contrasts. The bound may be wide with few agent cohorts. It does not address informative missingness, dependent runs, adaptive selection or new benchmark items. The implemented confirmatory gate requires a pre-collection registration record binding release, protocol, analysis and model identities, fixed equal human clusters per form, complete agent run cohorts, and original audit events. Incomplete or excluded clusters and withdrawals fail this strict design gate. Registration authenticity and independence are investigator attestations; software does not prove them. Other designs require separate justification. Bootstrap summaries are exploratory unless separately justified; hypothetical calibration does not establish the actual study's assumptions.

Eight judgments per shared-panel item requires 128 usable humans per condition at four items each. This establishes coverage, not power. Register sample, stopping and reserve rules after planning and before main outcomes.

RESULTS REQUIRED — TRIAL-PRIMARY

Insert matched-panel accuracy, denominators, independent clusters, justified uncertainty and registered contrasts. Separate full-pool scores.

RESULTS REQUIRED — TRIAL-SECONDARY

Insert obligation diagnostics, invalid/timeout/technical rates, capped timing and probes with denominators and applicability rules.

RESULTS REQUIRED — TRIAL-MODALITY

Report the registered probe extension with allocation checks, or explicitly state that it was not run.

RESULTS REQUIRED — TRIAL-ERRORS

Report error classes, counts, overlap rules and examples; distinguish observations from hypotheses about internal reasoning.

10 Limitations and extension boundary

Correct labels do not establish human ease, accessibility or construct validity. Public mechanisms, finite targets, construction filters and presentation choices constrain interpretation. General compilation can produce a large or visually complicated puzzle even when every intervention is preserved. Scrolling, zooming and focused rendering address access, but do not establish intuitive comprehension or eliminate navigation burden. The v4 target ambiguity demonstrates that execution and grading checks can pass while a display conveys the wrong goal. The v5 terminal projection and visual acceptance checks address that defect, but identifying the goal without technical instructions, comprehension and enjoyment still require a pilot of the corrected representation. Visual and serialized presentation remain different modalities. No human, live-model or infinite-strategy finding follows from software tests.

Removing fixed software caps does not remove computational cost. The explicit machine alphabet has 2^m valuations and the full visual forest scales with both state count and horizon. All-answer output may itself be exponential. A resource budget may end a run without deciding it. Each supported input is finite, and the theorems quantify over arbitrarily large finite inputs; no claim of a completed infinite computation is made. Neither a small-machine success nor a resource failure on one workstation determines what larger infrastructure can execute.

A strategy-necessity extension must supply the plant arena, owners, alternative moves, initial position, baseline policy/memory, editable commitments, background and effect monitors, and finite candidate language. The reviewed witness theorem then needs concrete monitor/solver correctness, reachable accepted witnesses, removal feasibility and exclusion certificates. One favorable lasso does not prove a universal policy guarantee. Remaining inside a winning region need not win, and a maximal SCC's minimum can miss losing subcycles. These are extension gates, not claims of this finite release.

RESULTS REQUIRED — TRIAL-LIMITATIONS

Update from the actual sample, missingness, ceiling/floor behavior, model constraints, presentation and unperformed controls.

11 Conclusion

TRACE specifies a finite explanation task with an exact backward-game interpretation, a general semantics-preserving Mealy-to-switch constructor and proved terminal display projection, a terminating minimal-answer algorithm and locally checkable evidence. Its intended comparison concerns people and AI using a common task, information and answer contract. Formal correctness and empirical benchmark validity remain distinct obligations.

RESULTS REQUIRED — TRIAL-CONCLUSION

Write conclusions from completed evidence, including null, reversed or ceiling findings. Do not predict a human advantage before measurement.

References

[1] Bernd Finkbeiner, Hadar Frenkel, Niklas Metzger, and Julian Siber. Synthesis of temporal causality. In *Computer Aided Verification (CAV)*, volume 14683 of *Lecture Notes in Computer Science*, pages 87–111. Springer, 2024. https://doi.org/10.1007/978-3-031-65633-0_5.

[2] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.

[3] Nikolaus Holzer, William Fishell, Baishakhi Ray, and Mark Santolucito. TempoBench: Reasoning execution without causal attribution is just simulation. *arXiv preprint arXiv:2510.27544*, 2026. Version 3, August 9, 2026. https://arxiv.org/abs/2510.27544v3.